

Brooke Macklin

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Summary

Results-driven **QA Lead / Quality Assurance Specialist** with 7+ years of experience in quality assurance, functional testing, and process improvement for enterprise and federal systems. Skilled in **manual, Application Programming Interface (API), regression testing**, User Acceptance Testing (UAT) coordination, and leading test teams while consulting with automation engineers to ensure scalable, high-quality solutions. Proven track record in **defect management, test documentation, and stakeholder collaboration** that improved pre-production defect detection by 40% and ensured 95%+ Service Level Agreement (SLA) compliance on support tickets.

Core Skills

QA & Testing Tools: Manual Testing, Regression, Integration, UAT, Zephyr Scale, Application Lifecycle Management (ALM)

Project & Collaboration Tools: Jira, Confluence, SharePoint, Microsoft Teams, Asana, Slack

Databases & Languages: Structured Query Language (SQL), Java, Python, HTML, XML, JSON, Excel

System Tools: Active Directory, Microsoft Office Suite

Methodologies: Waterfall, Agile/Scrum, Sprint Planning, Test Documentation, SDLC, Defect Tracking, Independent Verification & Validation (IV&V)

Professional Experience

Tyrula (FDA Contract) — McLean, VA | Aug 2022 – Oct 2024

Senior Quality Assurance Analyst

- Designed and executed 500+ functional, regression, and UAT test cases, improving defect detection by 40% pre-production.
- Conducted API testing and data audits, ensuring 100% accuracy for Food & Drug Administration (FDA) Human Resource and customer-facing systems.
- Coordinated with cross-functional Agile teams, reducing bug resolution time by 25%.
- Managed Tier 1–2 support tickets with 95% SLA compliance, enhancing customer satisfaction.
- Authored 10+ training materials, including guides, videos, & voiceovers, to streamline Quality Assurance (QA) onboarding and improve test process adoption.

Tech Stack: Jira, SharePoint, Java Query Language (JQL), API Testing, FDA Systems, Microsoft Suite, ActiveDirectory

OTOT Technologies (DHA & Navy) — Rosslyn, VA | Mar 2020 – Aug 2022

QA Engineer

- Directed IV&V and functional testing for Navy/Defense Health Agency (DHA) medical applications, ensuring 99% compliance with federal QA standards.
- Partnered with the automation team to identify areas for automation and directed test priorities to align with project needs.
- Executed API testing and managed defect triage in Jira, improving release quality and traceability.
- Facilitated multi-agency UAT cycles, aligning stakeholder expectations and validating critical requirements pre-release.
- Presented QA reports and quality metrics to executives for informed decision-making.

Tech Stack: Jira, Zephyr Scale, Confluence, ALM, API Testing, SQL, Microsoft Suite

Jester Music Co. — Hyattsville, MD | Jan 2018 – Jan 2020

UX Analyst & Digital Strategist

- Tested and optimized mobile/web User Experience (UX) across devices, ensuring Americans with Disabilities Act (ADA) compliance and accessibility standards.
- Improved Search Engine Optimization (SEO)/Search Engine Marketing (SEM) campaigns, increasing reach to 1M+ users and boosting streaming revenue 5x.
- Managed Content Management System (CMS) integrations and Secure Socket Layer (SSL) configurations, enhancing website security and performance.
- Designed User Interface (UI) flows, prototypes, and wireframes to guide cross-functional teams.
- Delivered analytics reports to stakeholders, driving informed UX improvements.

Tech Stack: CMS Platforms, SEO/SEM Tools, Google Analytics, UX Prototyping, Facebook Ads

Accenture Federal Services (CMS/Healthcare.gov) — Reston, VA | Oct 2015 – Oct 2017

Business & Systems Integration Analyst

- Authored Business Requirements Document (BRD)/Functional Requirements Document (FRD), executed 750+ test cases for Healthcare.gov modernization.
- Streamlined workflows, reducing process bottlenecks by 30% through Extensible Markup Language (XML) validation improvements.
- Coordinated requirement discovery sessions with stakeholders to align business and system goals.
- Conducted regression and functional testing, ensuring compliance with federal QA standards.
- Enhanced QA traceability documentation for long-term system maintenance.

Tech Stack: ALM, XML, SQL, Jira, JavaScript Object Notation (JSON), Agile Documentation, Healthcare.gov Systems

Education & Certification

B.S., Computer Science — Hampton University - May 2015

Certifications:

- Harvard Business School (HBS) Online: **Credential of Readiness (CORE) & Negotiation Mastery**
- SBS Swiss Business School: **International Finance**
- Berlin School of Economics & Law: **Entrepreneurship & Marketing**
- Université Grenoble: **French Language Studies**
- Scrum Alliance: **Certified ScrumMaster (CSM)**
- QAI Global: **Certified Software Tester (CSTE)**
- UIPath Academy: **RPA Developer Foundation Certification (UIPath)**

Additional Projects & Experience

Surveillance, Intelligence & IoT Integration Lab (2024/2025)

Designed and deployed a private security system integrating Eufy Closed-Circuit Television (CCTV) Surveillance Cameras, Abode Intrusion Detection Alarm System (IDS), and U-Tech Smart Intrusion Prevention Lock Systems (IPS), while deploying an encrypted private cloud server. Incorporated methodologies and technologies such as Redundancy (RAID), Virtual Private Networks (VPN's), Data Organization & Backup Strategy, and Alert Monitoring.

Robotic Process Automation & Music Publishing Lab (2024)

Designed and deployed RPA workflows using UiPath software to automate music publishing submissions from record labels. Techniques implemented resulted in reading in user-imputed data, entering website data with user-input data, and maintaining workflows with updates and enhancements. Consultation on automation-eligible processes was also conducted.

Cryptocurrency Graphics Processing Unit (GPU) Mining Lab (2021)

Built and maintained multiple multi-GPU mining rigs, installed & optimized Operating System (OS), firmware, and cooling systems. Troubleshoot OS, hardware, and network issues to maintain uptime. Consulted with cryptocurrency beginners on building their own GPU mining rigs. Maintained for 1+ year with minimal issues.

Cyber Aquatic Systems Lab (2013)

Built a Graphical User Interface (GUI) for Aquatic Underwater Vehicles (AUVs) and migrated Windows tools to Linux, applying accessibility-first design as part of a summer research internship conducted at the University of Connecticut.

Multi-Robot Systems Lab (2012)

Developed a robotics curriculum at Rice University in Python scripting language for high school students, promoting inclusive tech education, as part of a summer research internship. Participating students were given quizzes, homework, practicals, survey's and a final exam to test student growth throughout the course. This research project won 2nd place at the 2013 Association of Computer Science Departments at Minority Institutions (ADMI) Research Conference.

Network Operations Administrator Lab (2010)

Managed and refurbished 30+ Windows/Linux systems for re-distribution, ensuring network security and tech accessibility for students throughout Montgomery County Public Schools (MCPS). Competed and won 3rd place in SkillsUSA Computer Technician I.

Key Achievements

- Reduced defect leakage by 40% across multiple projects through robust regression testing strategies.
- Consulted with automation engineers to define automation priorities and timelines, ensuring alignment with QA strategies.
- Improved onboarding and QA consistency by developing 10+ training manuals for testers and stakeholders.
- Delivered 1M+ user reach through UX testing and SEO improvements, boosting revenue streams 5x.
- Achieved 95%+ SLA compliance in support ticket resolution, strengthening customer satisfaction.

Professional Closure

A proven QA Lead and Quality Assurance Specialist with a strong blend of technical expertise, process optimization, and leadership skills, consistently delivering high-quality, compliant, and scalable systems. Ready to contribute to organizations seeking excellence in software testing and quality delivery.